

Task 1: Adversity Profile of Clients

For each client and horizon $\tau \in \{5, 10, 15, 20, 25, 30\}$, a trade is classified as **adverse** if the LP's PnL is negative when the position is assumed to be closed at time τ . Since volume is always positive, the adverse-trade condition reduces to:

$$\text{side} \cdot (M_\tau - TP) < 0.$$

The *adversity profile* is the percentage of a client's trades satisfying this condition at each horizon.

Client	$\tau = 5$	$\tau = 10$	$\tau = 15$	$\tau = 20$	$\tau = 25$	$\tau = 30$
A	39.7	41.4	42.2	42.0	41.9	41.7
B	40.8	42.4	43.2	43.3	43.4	43.3
C	41.4	43.3	44.2	44.8	45.4	45.9
D	43.3	46.5	48.2	49.7	50.6	51.8
E	45.0	48.5	50.9	52.1	53.8	54.8
F	47.0	51.7	54.3	56.9	59.5	61.9

The results reveal two clear patterns.

Adversity increases with horizon τ . The longer the LP holds the position before closing, the more likely the trade becomes loss-making. The increase is mild for Clients A and B (adversity stays in the 40–43% range across all τ) but is substantially stronger for Clients D, E, and F.

Significant cross-client heterogeneity. Clients A and B have the least adverse flow, remaining below 44% even at $\tau = 30$ s. Client C shows a moderate increase (41.4% $\rightarrow$ 45.9%). Clients D, E, and F are markedly more adverse: Client D crosses 50% by $\tau = 25$ s; Client E reaches 54.8% at $\tau = 30$ s; and Client F is the most adverse, rising from 47.0% at $\tau = 5$ s to 61.9% at $\tau = 30$ s — the steepest ascent of all six clients.

This pattern suggests that Clients E and F exhibit informed or *toxic* flow: their trades are systematically followed by mid-price moves that go against the LP over longer horizons. From a market-making perspective, the LP should quote wider spreads or externalize a larger fraction of flow from these clients. Clients A and B are relatively benign and can support more competitive quoting.

A limitation is that the adversity profile captures only the *frequency* of adverse trades, not their magnitude. A client with fewer but larger adverse trades can still be net-costly; therefore this profile should be read alongside the expected-PnL calculations in Task 2.

